

NARA, Mali

WMO: 612330

Lat: 15.167N

Lon: 7.283W

Elev: 869

StdP: 14.24

Time Zone: 0.00 (E00)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.3	58.8	13.1	11.1	84.2	16.7	13.2	84.7	13.8	82.0	12.4	81.2	1.6	40	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.8	109.7	69.5	108.2	69.5	106.7	69.5	79.9	90.7	79.1	89.8	78.4	89.1	5.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	147.4	84.1	76.6	143.3	83.2	76.0	140.2	83.0	44.2	90.6	43.3	89.8	42.6	89.1	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.0	9.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.1	74.0	78.9	87.1	92.0	95.4	93.9	86.1	82.9	85.2	87.8	81.9	75.5
	DBStd	7.73	3.88	5.93	5.27	4.23	3.82	4.13	4.07	3.21	3.50	2.57	3.31	4.55
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	3	2	0	0	0	0	0	0	0	0	0	0	1
	CDD50	12803	744	809	1149	1261	1406	1318	1118	1021	1056	1172	957	792
	CDD65	7330	281	389	684	811	941	868	653	556	606	707	507	327
	CDH74	104734	2979	4820	10306	13570	16126	14232	8812	6276	7458	9941	6618	3596
	CDH80	65157	1276	2675	6679	9504	11736	9977	4844	2845	3908	6089	3883	1741
Wind	WSAvg	4.1	4.2	4.9	5.3	5.0	4.6	5.0	4.1	3.2	2.8	3.0	3.1	4.1
Precipitation	PrecAvg	15.5	0.0	0.0	0.0	0.1	0.4	1.6	4.5	5.5	2.8	0.6	0.0	0.0
	PrecMax	23.5	0.1	0.1	0.4	1.2	2.2	4.5	8.3	9.6	7.4	2.2	0.5	0.1
	PrecMin	7.8	0.0	0.0	0.0	0.0	0.0	0.0	1.4	1.5	0.6	0.0	0.0	0.0
	PrecStd	3.6	0.0	0.0	0.1	0.2	0.5	1.1	1.6	2.1	1.4	0.6	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	104.2	107.6	110.6	112.1	111.2	104.2	98.9	103.0	104.9	101.6	97.8
		MCWB	60.2	65.9	64.9	67.5	69.7	70.1	75.8	77.2	75.0	69.5	65.1	63.2
	2%	DB	92.1	100.1	105.3	108.6	110.1	108.9	100.3	95.3	99.6	103.1	99.8	95.0
		MCWB	59.3	63.3	65.5	67.6	70.2	70.6	75.5	77.8	76.4	69.0	64.6	62.3
	5%	DB	89.8	97.2	103.3	107.1	108.6	107.1	97.7	93.3	97.0	101.7	98.3	92.6
		MCWB	58.3	61.7	64.1	67.5	70.3	71.7	75.8	77.7	77.3	69.1	64.3	61.6
	10%	DB	87.2	93.9	101.3	105.3	106.9	104.9	95.6	91.5	94.6	100.2	96.3	89.6
		MCWB	57.4	60.2	63.5	67.0	70.4	72.5	76.0	77.5	77.7	69.5	64.4	60.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	68.3	73.6	74.8	77.5	78.7	80.6	80.9	81.4	79.2	72.4	66.7
		MCDB	86.8	99.3	100.2	99.3	97.3	91.3	89.4	91.1	93.4	93.4	90.3	93.1
	2%	WB	61.4	64.7	69.4	72.9	75.9	77.1	78.9	79.5	80.1	77.3	69.8	64.2
		MCDB	85.1	96.2	94.4	100.5	96.7	94.4	89.6	89.3	91.2	90.9	88.3	91.8
	5%	WB	59.5	62.8	66.5	71.7	75.1	76.2	78.1	78.9	79.2	76.2	67.4	62.3
		MCDB	86.8	94.0	100.0	99.6	96.8	94.3	89.4	88.8	90.2	90.0	89.4	89.9
	10%	WB	58.1	61.0	64.5	70.3	74.2	75.4	77.5	78.3	78.5	75.2	65.4	60.7
		MCDB	85.8	91.3	98.4	98.7	96.9	94.1	89.2	87.7	89.3	89.3	91.4	87.9
Mean Daily Temperature Range	5% DB	MDBR	27.3	27.3	27.5	26.2	22.8	20.5	16.7	15.7	19.1	26.0	29.5	27.7
		MCDBR	31.1	30.6	28.2	26.7	24.8	22.5	19.3	18.0	22.1	28.2	31.8	31.2
		MCWBR	12.2	11.2	10.2	8.7	8.2	6.4	4.8	5.1	6.0	7.4	10.5	11.9
	5% WB	MCDBR	29.2	28.4	26.6	21.8	21.0	20.0	16.6	16.6	19.1	22.5	28.2	29.9
		MCWBR	12.0	11.2	10.6	8.5	7.3	6.1	5.7	5.9	7.0	6.3	10.3	12.1
Clear-Sky Solar Irradiance	taub		0.506	0.532	0.643	0.657	0.699	0.720	0.651	0.593	0.581	0.572	0.494	0.473
	taud		1.862	1.791	1.573	1.596	1.537	1.550	1.712	1.849	1.855	1.831	1.923	1.957
	Ebn at Noon		248	248	226	223	210	203	219	234	237	235	249	253
	Edh at Noon		62	70	89	87	91	89	76	67	66	66	58	55
All-Sky Solar Radiation	RadAvg		1674	1879	2084	2185	2083	1980	1901	1879	1929	1872	1734	1628
	RadStd		104	101	98	85	108	62	66	57	78	46	66	66

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page